

Candidate Seat No.:_____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (M.C.A.)

Semester-I University Examination March 2018

CA722 DATABASE TECHNOLOGIES

Date: 27/03/2018, Tuesday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

Instructions:

- 1. Attempt the two sections in separate answer books.**
- 2. Figure to the right indicate marks.**
- 3. Make suitable assumptions wherever necessary and state them.**

SECTION I

Q-1 Do as directed.

- (A)** Differentiate Data and Information. **[01]**
- (B)** DDL stands for _____. **[01]**
- (C)** NULL is allowed in primary key field. [True/False] **[01]**
- (D)** Define: Entity and Attribute. **[02]**
- (E)** Construct an E-R diagram for the Online Shopping System. Write down all assumptions that you make about the mapping constraints. **[06]**

Q-2 Answer following questions.

- (A)** Explain *Process per DBMS worker* Process Model with suitable diagram. **[06]**
- (B)** List and explain advantages of Database Management System. **[06]**

OR

Q-2 Answer following questions.

- (A)** What is normalization? Explain 1NF, 2NF and 3NF with suitable example. **[06]**
- (B)** What is Query Rewriter? Explain View Expansion in Query Rewriter. **[06]**

Q-3 Answer following questions. [Attempt any three]

[12]

- (A)** List and explain different types of database user.
- (B)** Explain the Selection and Projection operations of Relational Algebra with suitable example.
- (C)** Explain NUMA architecture in detail.
- (D)** Explain multilevel indexing with an example.
- (E)** What is *Shared Disk* parallel architecture? Explain it with diagram.

SECTION II

Q-4 Do as directed.

(A) What is Schedule? Explain with an example. [02]

(B) What is the purpose of Database Recovery? [02]

(C) Table: Employee [07]

EmpId	EmpName	EmpSalary	EmpCity
E001	Suresh	25000	Anand
E002	Ramesh	35000	Gandhinagar
E003	Paresh	40000	Anand

Write a SQL query for following:

- Create above mention table using create command.
- Display all information of employee whose EmpId is E002.
- Display information of all employees whose name start with 'R'.
- Update the salary of all employee by 10%.
- Delete all employees who live in 'Gandhinagar'.
- Display the highest salary paid to the employee.
- Add the column EmpBirthDate in table.

Q-5 Answer following questions.

(A) Discuss Sort-Merge Join algorithm with an example. [08]

(B) Define: Atomicity and Consistency [04]

OR

Q-5 Answer following questions.

(A) Discuss Hybrid Hash-Join algorithm with an example. [08]

(B) Define: Isolation and Durability [04]

Q-6 Answer following questions.

(A) Explain three main principles used in ARIES recovery algorithm with an example. [06]

(B) Differentiate where clause and having clause in SQL with an example. [06]

OR

Q-6 Answer following questions.

(A) What is Transaction Log? Explain about the types of actions recorded in the Transaction Log. [06]

(B) Explain left join and right join in SQL with an example. [06]